

PyTorch with Slurm and clusters

From a training script to a job on a GPU

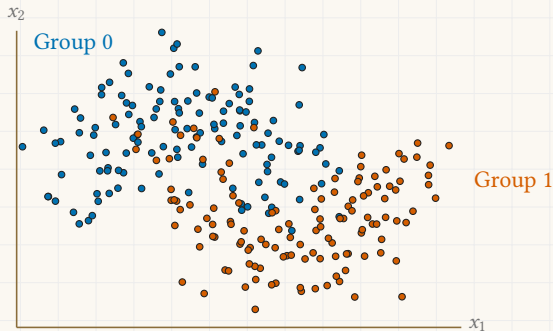
Carlos Cotrini

ETH Zurich

6 October 2026

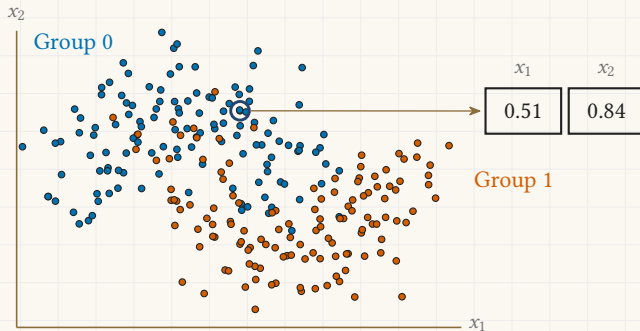
Two crescents, one network

Each point has two coordinates and belongs to one of two groups. The rest of the hour shows how to train a network that tells them apart, on a GPU of Santis.



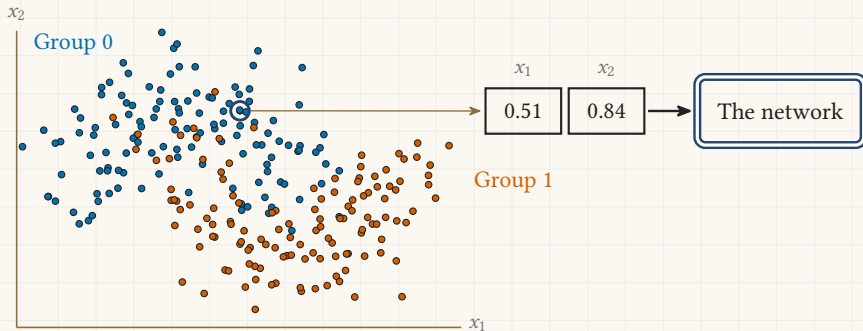
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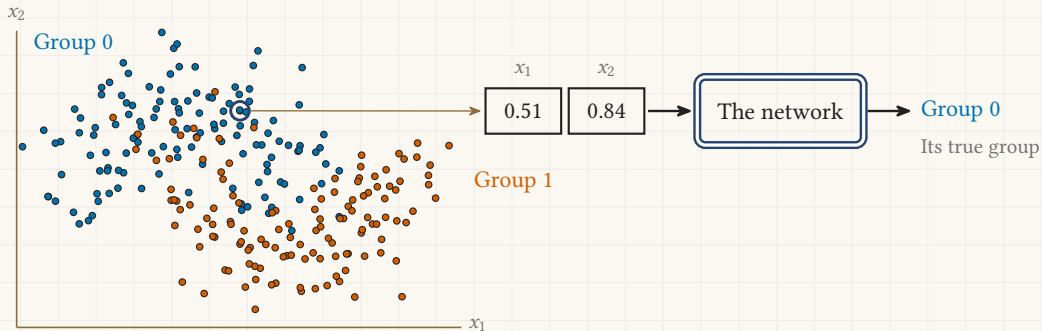
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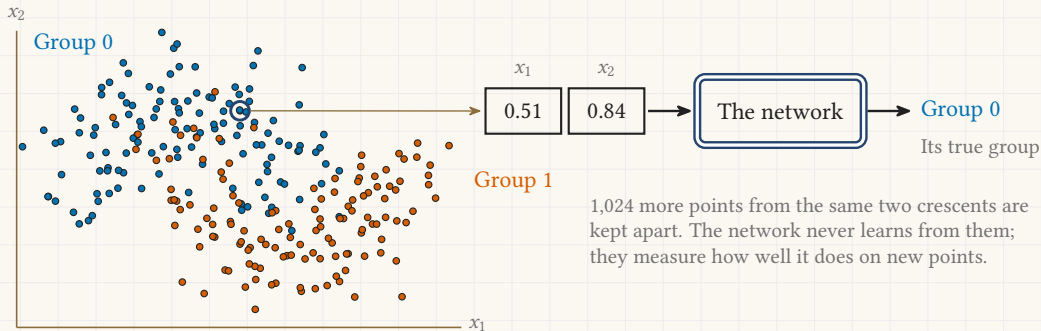
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The roadmap

Two runs of one PyTorch file on a GPU of Santis, without and then with weight decay, in four steps.

1

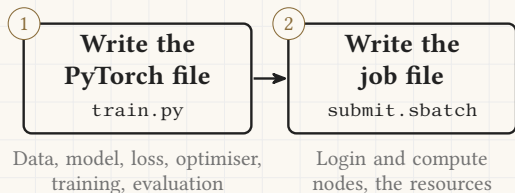
**Write the
PyTorch file**

`train.py`

Data, model, loss, optimiser,
training, evaluation

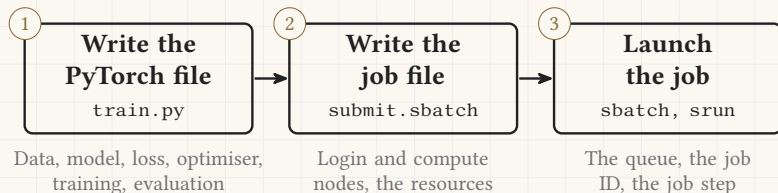
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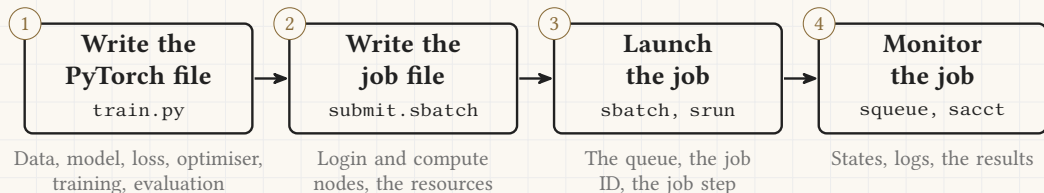
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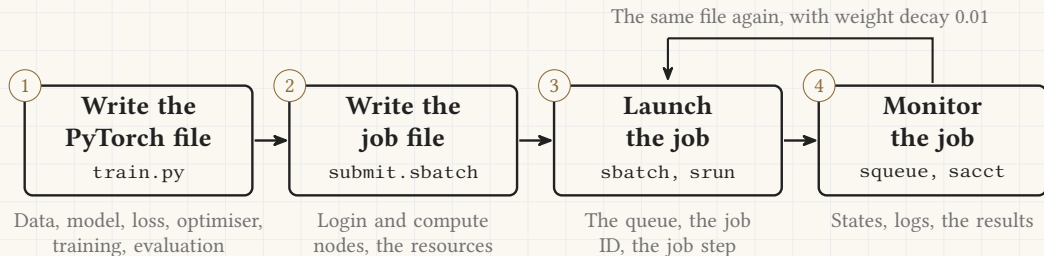
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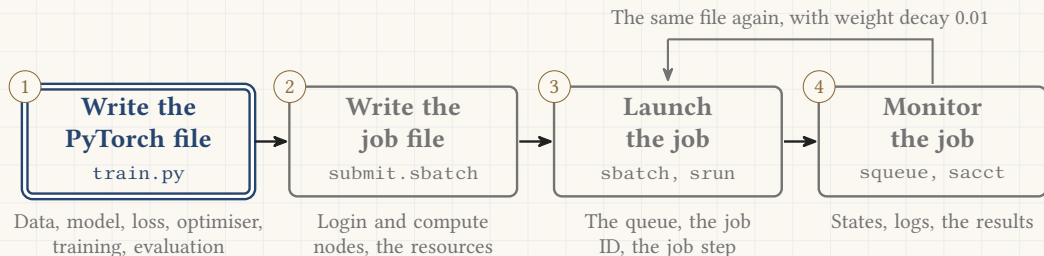
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The roadmap

Two runs of one PyTorch file on a GPU of Santis, without and then with weight decay, in four steps.



Four objects, and what each one does

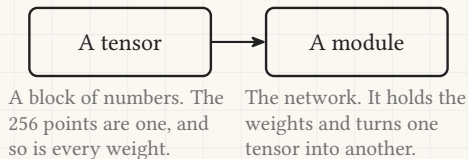
This morning's loop, as four PyTorch objects. `train.py` builds each of them, then runs the loop.

A tensor

A block of numbers. The 256 points are one, and so is every weight.

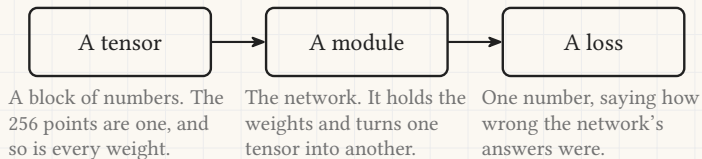
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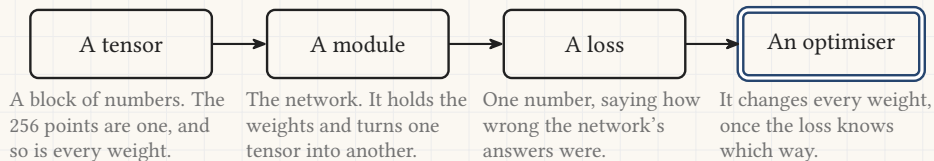
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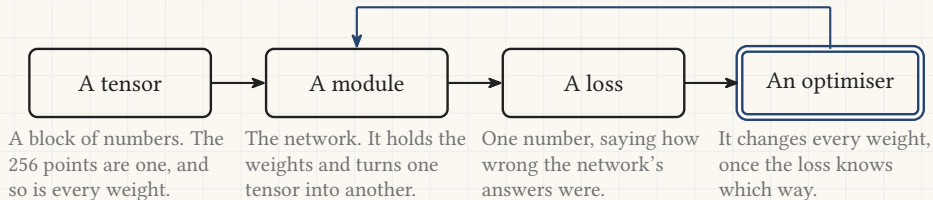
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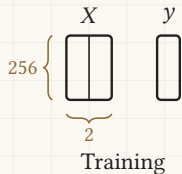
Round and round, 160 times. That is training.



The data, as two tensors

The first object. One call makes both sets, from the same two crescents.

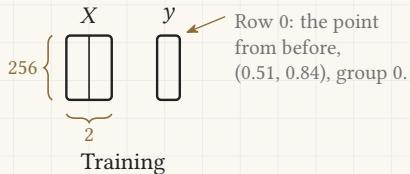
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train_data, val_data = make_data()
```



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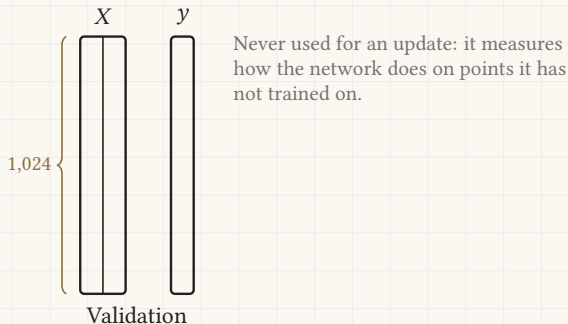
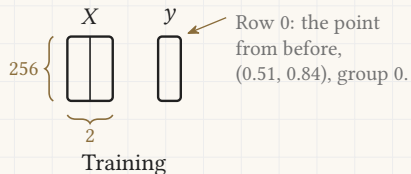
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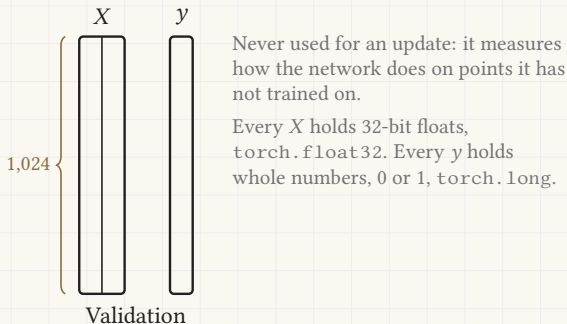
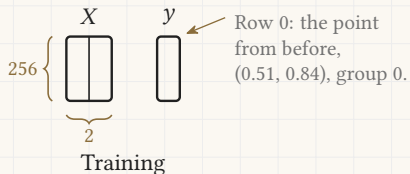
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Minibatches of 64

This morning, one random house per round. Here, 64 random points per round.

```
train_loader = DataLoader(TensorDataset(*train_data), batch_size=args.batch_size,  
                           shuffle=True, generator=generator, num_workers=0)
```



X, 256 rows

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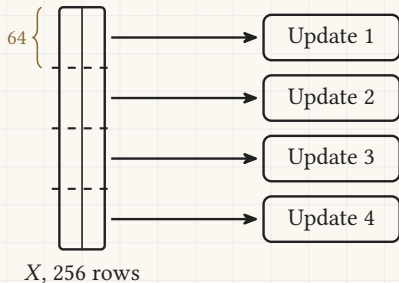


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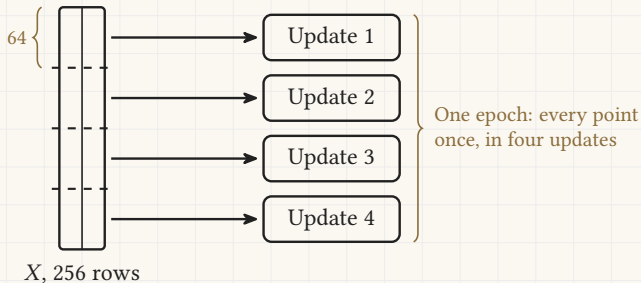
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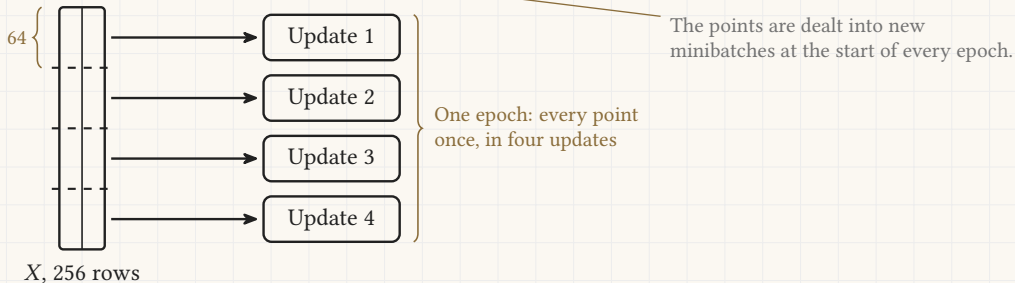
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```



The network, as blocks

The second object. One minibatch in, two scores per point out. Drawn at width 16 to fit on the slide; both runs use width 128.

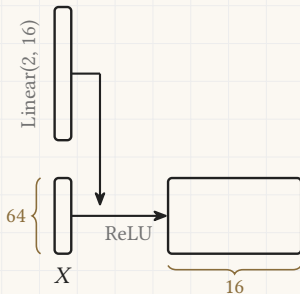
```
def make_model(width):  
    return nn.Sequential(nn.Linear(2, width), nn.ReLU(),  
                          nn.Linear(width, width), nn.ReLU(), nn.Linear(width, 2))
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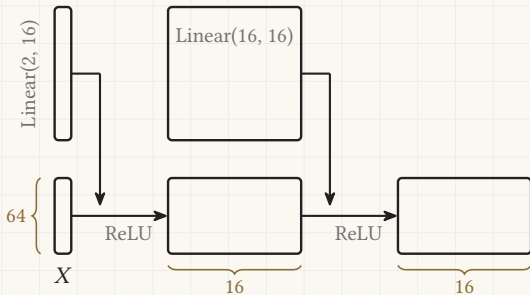
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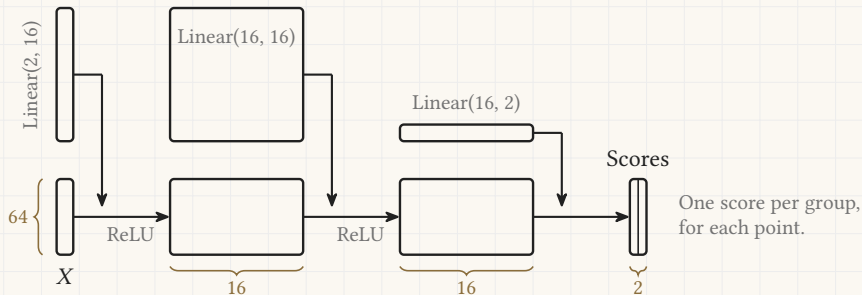
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The numbers this network holds

Every number the optimiser will change, at width 16 as drawn and at width 128 as run, to one scale.



Width 16

The numbers this network holds

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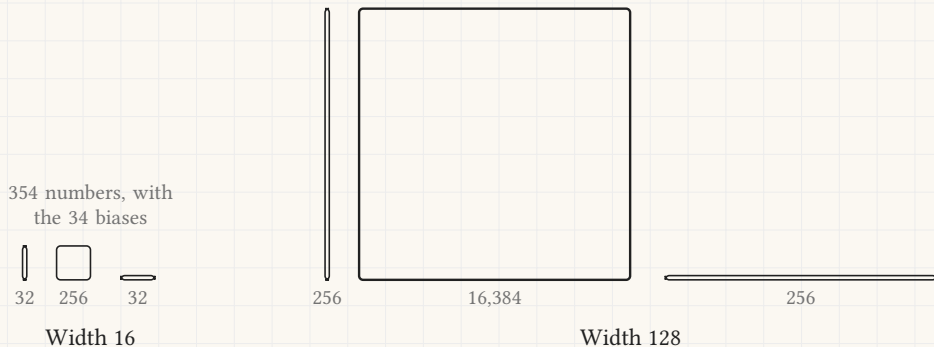
354 numbers, with
the 34 biases



Width 16

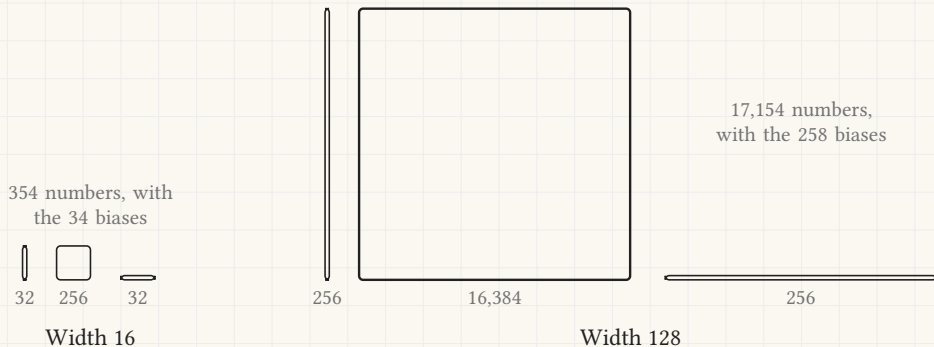
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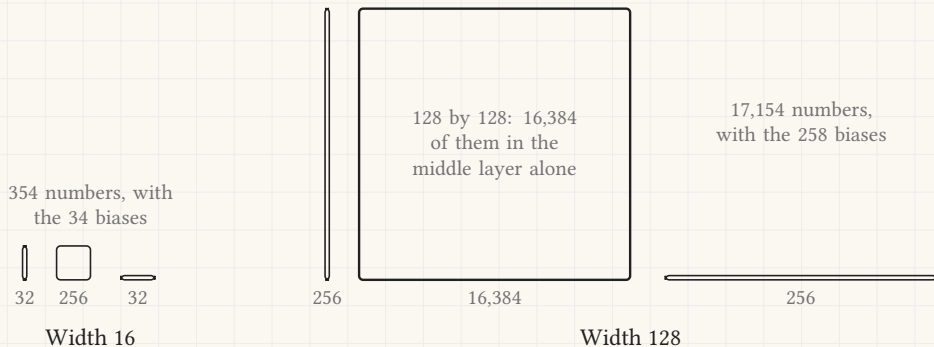
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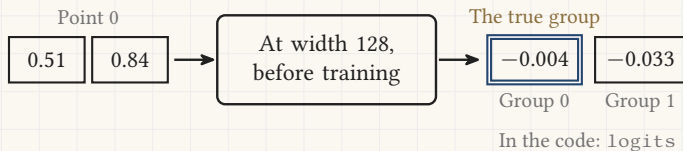
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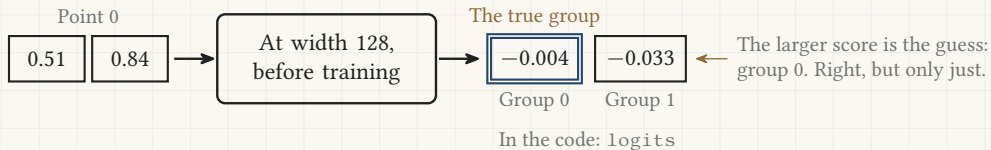
Two scores out, one verdict

The third object. This morning's verdict was a squared miss, `nn.MSELoss()`. Here the network gives one score per group.



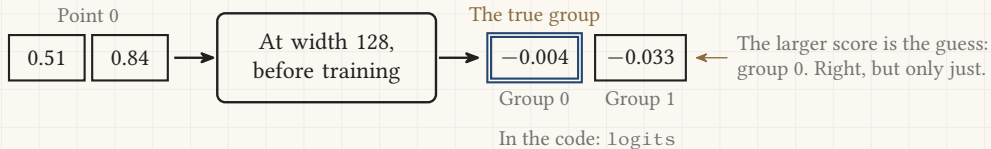
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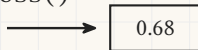
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loss_fn = nn.CrossEntropyLoss()
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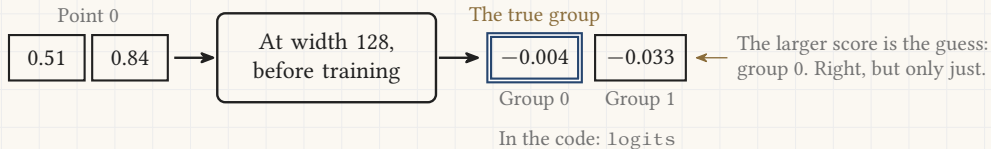
```
loss = loss_fn(logits, y)
```



The loss for point 0

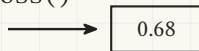
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The loss for point 0

Large unless the true group's score is far ahead. Here it is only just ahead. Training moves the weights to make the loss smaller.

The loss and the optimiser

The fourth object, and the two lines of `train.py` that create the last two.

```
optimizer = torch.optim.Adam(model.parameters(), lr=args.lr,  
                              weight_decay=args.weight_decay)  
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The loss and the optimiser

The fourth object, and the two lines of `train.py` that create the last two.

The network's 17,154
numbers, at width 128



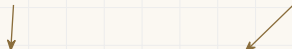
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This morning's α , now
scaling every step: 0.01
by default



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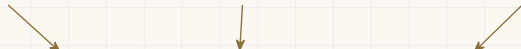
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The fourth object, and the two lines of `train.py` that create the last two.

Adam: each number gets a
step size of its own, tuned
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optimizer = torch.optim.Adam(model.parameters(), lr=args.lr,  
                             weight_decay=args.weight_decay)
```

```
loss_fn = nn.CrossEntropyLoss()
```

Weight decay: every step also pulls each number a little toward zero. 0.0 by default; the second run sets 0.01

One training step

The four objects in one loop, one minibatch at a time: the function `training_step`.

Minibatch

`x, y = batch`

One training step

The four objects in one loop, one minibatch at a time: the function `training_step`.

Minibatch

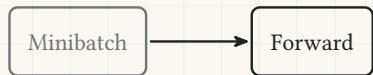
Backward
adds to old
gradients,
so clear
them first.

```
x, y = batch
```

```
optimizer.zero_grad()
```

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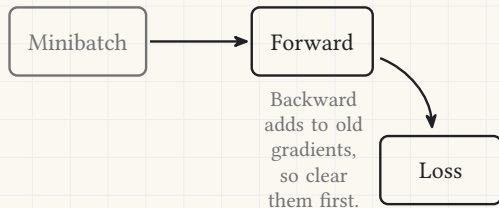
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x, y = batch
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```
optimizer.zero_grad()  
logits = model(x)
```

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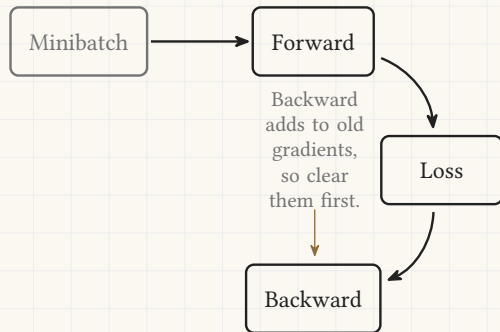
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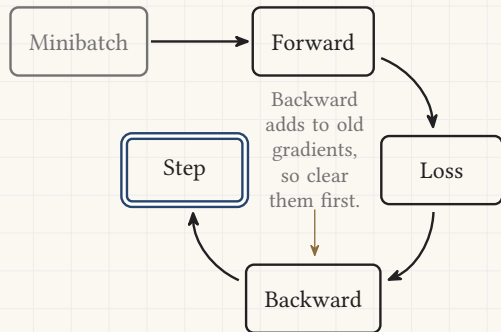


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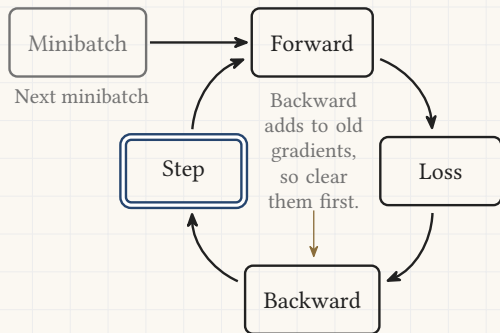


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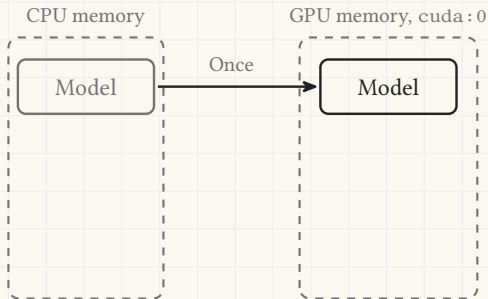


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Model and minibatch on the same device

The same step, with the line that moves each minibatch to the GPU, where the model already is.



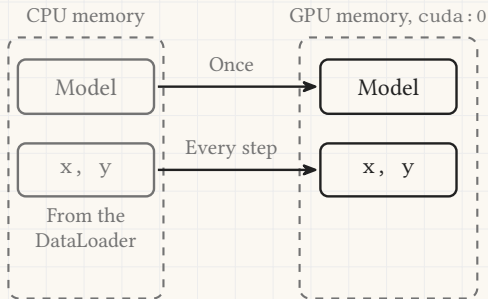
```
x, y = batch
x, y = x.to(device), y.to(device)
optimizer.zero_grad()
logits = model(x)
loss = loss_fn(logits, y)
loss.backward()
optimizer.step()
```

`model = make_model(args.width).to(device)`

In `train.py`, once, before the optimiser is built

Model and minibatch on the same device

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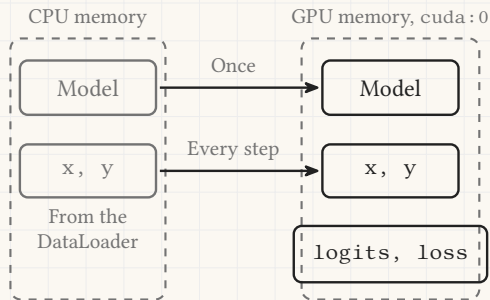
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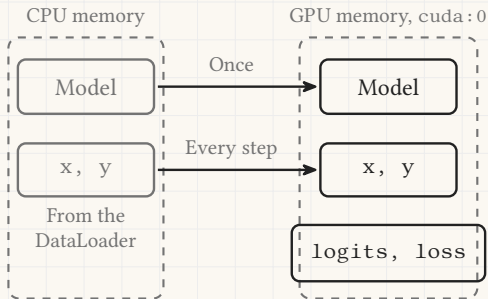
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Model and minibatch on the same device

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`train.py` picks `cuda:0` only inside a job, with exactly one GPU visible.

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x, y = batch
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optimizer.zero_grad()
logits = model(x)
loss = loss_fn(logits, y)
loss.backward()
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```
model = make_model(args.width).to(device)
```

In `train.py`, once, before the optimiser is built

Forty epochs of four steps

The loop around the step. After every epoch, both sets are measured with the same weights.

```
for epoch in range(1, args.epochs + 1):  
    model.train()  
    for batch in train_loader:  
        loss, evidence = training_step(model, batch, optimizer, loss_fn, device)
```

Epoch 1



Epoch 2



...

Epoch 40



Forty epochs of four steps

The loop around the step. After every epoch, both sets are measured with the same weights.

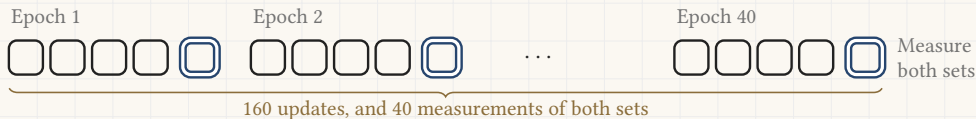
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for epoch in range(1, args.epochs + 1):  
    model.train()  
    for batch in train_loader:  
        loss, evidence = training_step(model, batch, optimizer, loss_fn, device)  
    tr_loss, tr_acc = evaluate(model, train_eval, device)  
    va_loss, va_acc = evaluate(model, val_loader, device)
```



Forty epochs of four steps

The loop around the step. After every epoch, both sets are measured with the same weights.

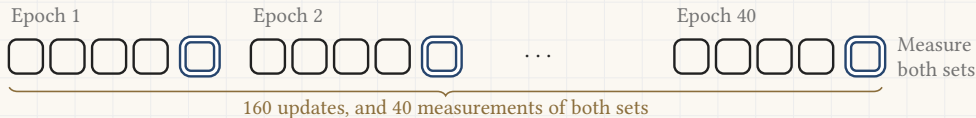
```
for epoch in range(1, args.epochs + 1):  
    model.train()  
    for batch in train_loader:  
        loss, evidence = training_step(model, batch, optimizer, loss_fn, device)  
    tr_loss, tr_acc = evaluate(model, train_eval, device)  
    va_loss, va_acc = evaluate(model, val_loader, device)
```



Forty epochs of four steps

The loop around the step. After every epoch, both sets are measured with the same weights.

```
for epoch in range(1, args.epochs + 1):  
    model.train() ← Back to training behaviour at the start of every epoch  
    for batch in train_loader:  
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```



Measuring without learning

`evaluate()` runs the network over a whole set and returns its loss and accuracy, without changing a weight.

```
@torch.no_grad()
def evaluate(model, loader, device):
    model.eval()
    total_loss, correct, count = 0.0, 0, 0
    for x, y in loader:
        x, y = x.to(device), y.to(device)
        logits = model(x)
        total_loss += nn.functional.cross_entropy(logits, y, reduction="sum").item()
        correct += (logits.argmax(1) == y).sum().item()
        count += len(y)
    return total_loss / count, correct / count
```

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`@torch.no_grad()`



No computation graph is built, which saves memory and time

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@torch.no_grad()  ← No computation graph is built, which saves
def evaluate(model, loader, device):                      memory and time
    model.eval()    ← Evaluation behaviour for every layer; none
                    of this network's layers changes with it
    total_loss, correct, count = 0.0, 0, 0
    for x, y in loader:
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← Evaluation behaviour for every layer; none of this network's layers changes with it

← Correct when the larger score belongs to the true group

Measuring without learning

`evaluate()` runs the network over a whole set and returns its loss and accuracy, without changing a weight.

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```

← No computation graph is built, which saves memory and time

← Evaluation behaviour for every layer; none of this network's layers changes with it

← Correct when the larger score belongs to the true group

No optimizer.step() anywhere: evaluation only reads

Layer behaviour and gradient tracking are separate

The two switches of the last slide do different jobs, and evaluation code uses both.

```
model.eval()
```

*Dropout turns off; BatchNorm
switches to its running statistics*

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Stops autograd building the computation graph

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Dropout turns off; BatchNorm switches to its running statistics

Says nothing about gradients

`torch.no_grad()`

Stops autograd building the computation graph

Saves the memory and compute gradients cost

The file takes its settings as flags

The file is complete. What the two runs change arrives on the command line.

```
p.add_argument("--width", type=int, default=128)
p.add_argument("--epochs", type=int, default=40)
p.add_argument("--batch-size", type=int, default=64)
p.add_argument("--lr", type=float, default=0.01)
p.add_argument("--weight-decay", type=float, default=0.0)
p.add_argument("--seed", type=int, default=42)
p.add_argument("--output", type=Path, required=True)
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Left at their defaults, the same in both runs

← The one setting the two runs change: 0.0, then 0.01

← A new folder for every run; an existing one is refused

In it:

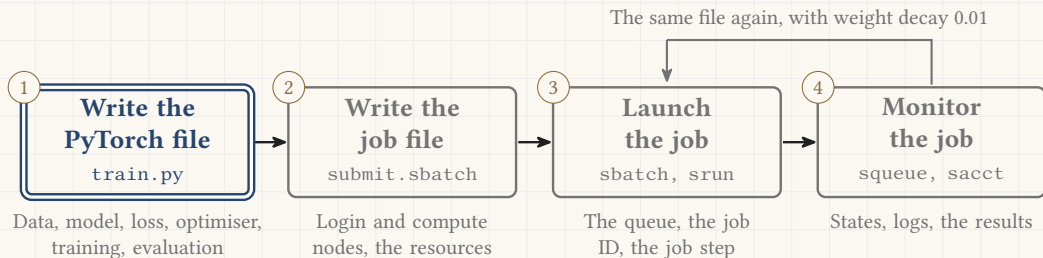
config.json

metrics.csv

summary.json

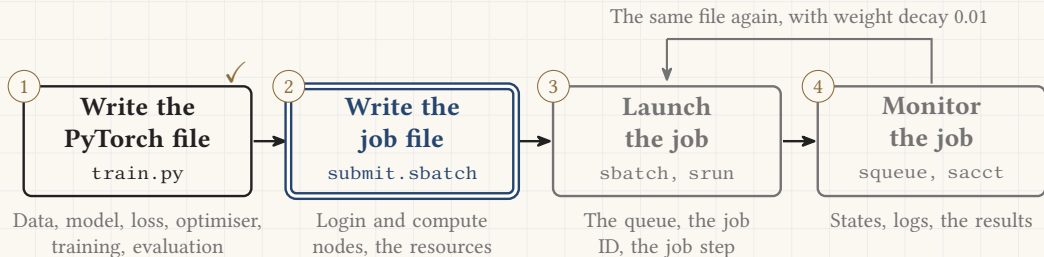
Step 2 of 4

Write the job file



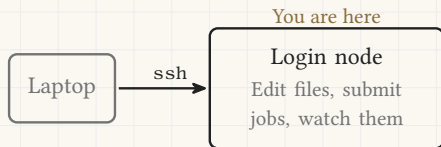
Step 2 of 4

Write the job file



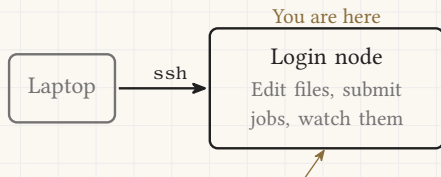
Login node and compute nodes

Where you log in is not where you train, and `train.py` knows it.



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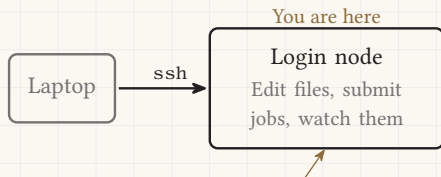


Started outside a job, `train.py` refuses to train:

No Slurm allocation. Submit `submit.sbatch`; do not train on a login node.

Login node and compute nodes

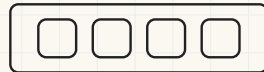
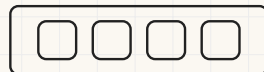
Where you log in is not where you train, and `train.py` knows it.



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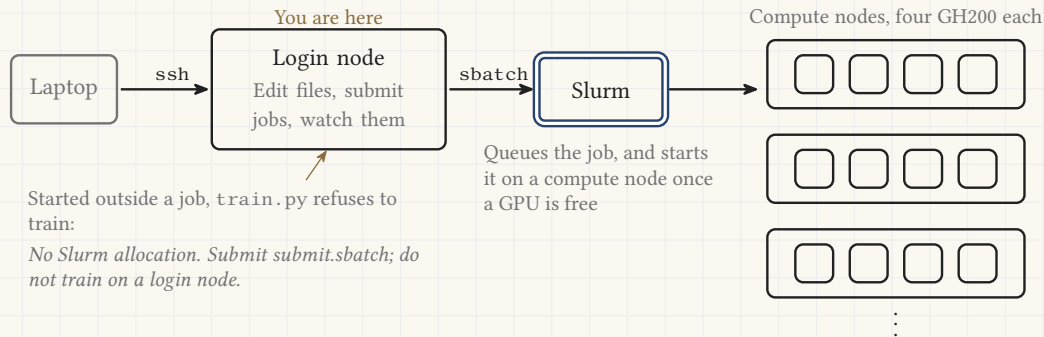
Compute nodes, four GH200 each



⋮

Login node and compute nodes

Where you log in is not where you train, and `train.py` knows it.



The job file asks for resources

The top of `submit.sbatch`: requests that Slurm reads before anything runs.

```
submit.sbatch
#!/bin/bash
#SBATCH --job-name=cx-pytorch
#SBATCH --account=ai-tutorial-course2026-cscs
#SBATCH --partition=normal
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --gpus-per-task=1
#SBATCH --cpus-per-task=1
#SBATCH --time=00:05:00
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What the file asks for

Paid by the course account, queued in the normal partition

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One GH200

One core

Without a GPU request: one core, no GPU, and `train.py` stops

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PyTorch from a CSCS software image, a uenv
Log names: %x the job name, %j its number

The job file then runs the script

Below the requests, ordinary shell lines. Slurm runs them on the compute node, once the job starts.

```
submit.sbatch, below the requests
cd "$SLURM_SUBMIT_DIR"
width="${1:-128}"
weight_decay="${2:-0.0}"
output="runs/${SLURM_JOB_ID}"
srun python -u train.py \
    --width "$width" --weight-decay "$weight_decay" --output "$output"
```

The job file then runs the script

Below the requests, ordinary shell lines. Slurm runs them on the compute node, once the job starts.

```
ooo
```

```
$ job1=$(sbatch --parsable submit.sbatch 128 0.0)
```

submit.sbatch, below the requests

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cd "$SLURM_SUBMIT_DIR"
```

```
width="${1:-128}"
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```
weight_decay="${2:-0.0}" ← The second argument after the file name: 0.0, and later 0.01
```

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```

← Set by Slurm: one folder per job

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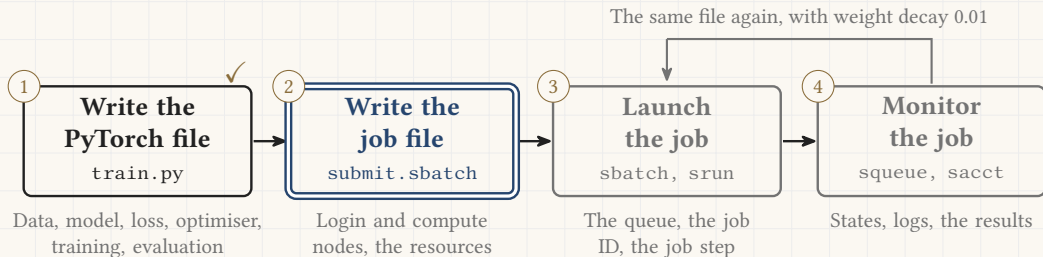
```
srun python -u train.py \
```

← Starts train.py inside the allocation, as a job step

```
--width "$width" --weight-decay "$weight_decay" --output "$output"
```

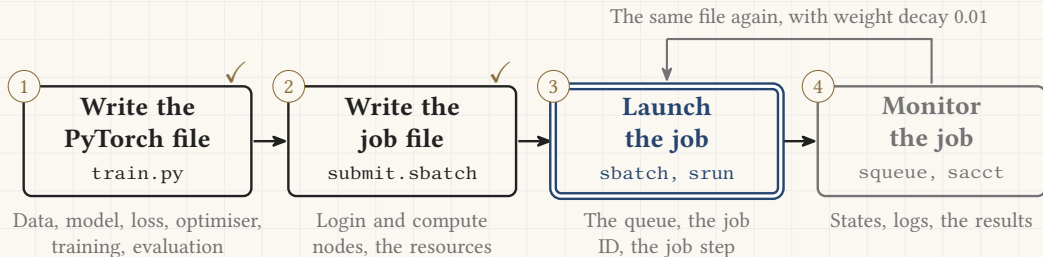
Step 3 of 4

Launch the job



Step 3 of 4

Launch the job



sbatch puts the job in the queue

Typed on the login node. It returns at once, with the job's number.

```
ooo
```

```
$ mkdir -p logs
```



First: Slurm opens the log files
before the script runs, so without
logs/ the job leaves no log at all

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```
$ mkdir -p logs  
$ job1=$(sbatch --parsable submit.sbatch 128 0.0)  
$ job1=${job1%%;*}    Keeps the number alone  
$ printf 'Baseline job ID: %s\n' "$job1"  
Baseline job ID: <JOB_ID>
```

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The queue



Your job



PENDING: waiting for a free GPU



Towards a free GPU

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Baseline job ID: <JOB_ID>
```

First: Slurm opens the log files before the script runs, so without logs/ the job leaves no log at all

A job ID confirms submission, not completion.

The queue



Your job



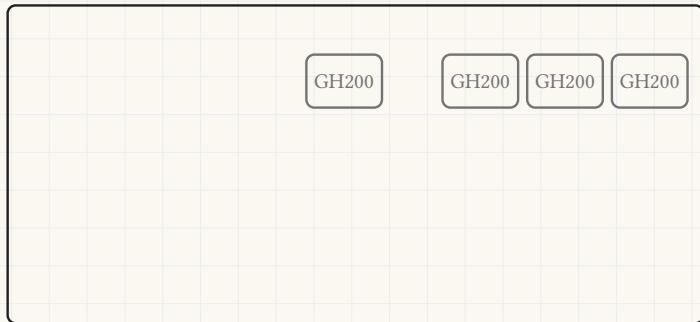
PENDING: waiting for a free GPU



Towards a free GPU

srun starts the training

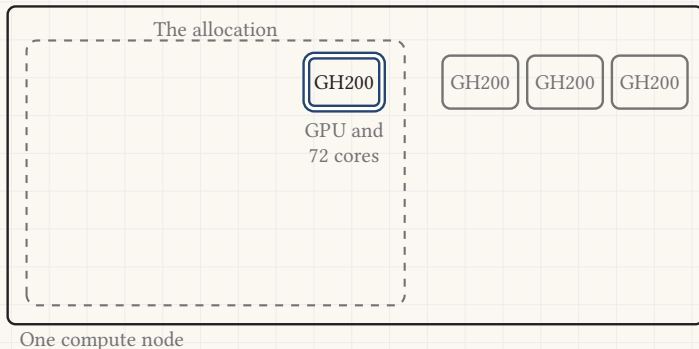
When a GPU is free, Slurm runs `submit.sbatch` on a compute node, and its `srun` line starts `train.py`.



One compute node

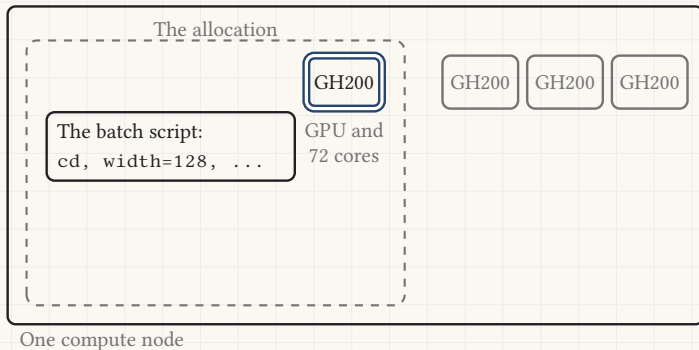
srun starts the training

When a GPU is free, Slurm runs `submit.sbatch` on a compute node, and its `srun` line starts `train.py`.



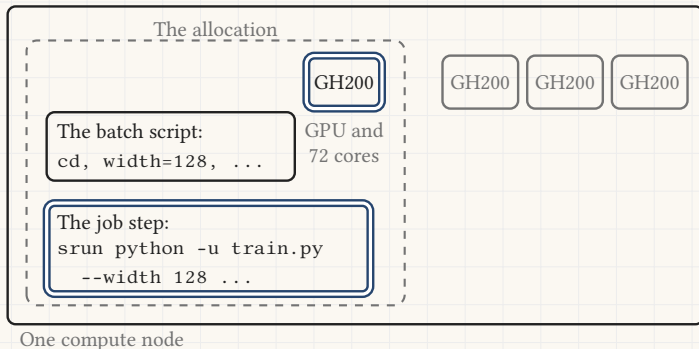
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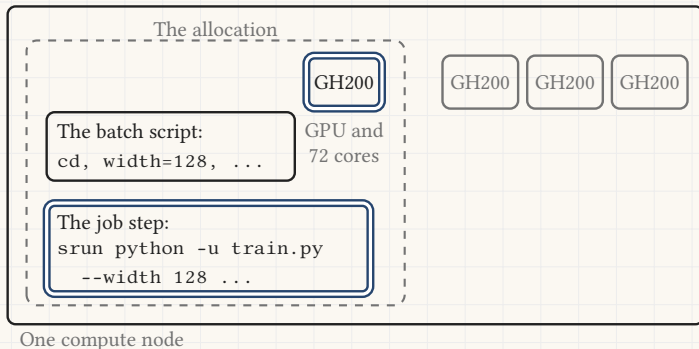
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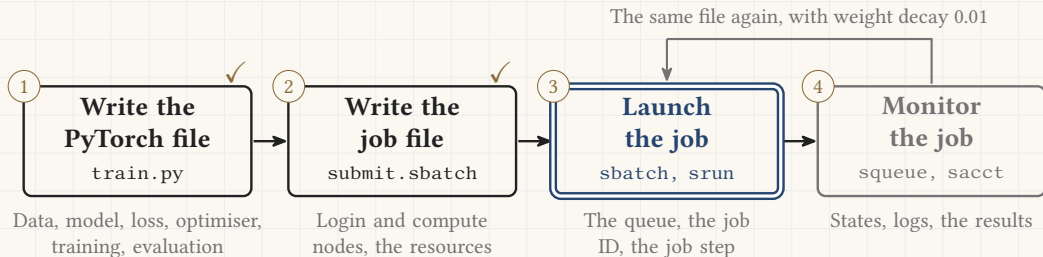
Inside the job, the checks of `train.py` pass:

- ✓ `SLURM_JOB_ID` is set: this is an allocation
- ✓ `CUDA` works and exactly one GPU is visible: the device is `cuda:0`

So the 160 updates run on the GH200.

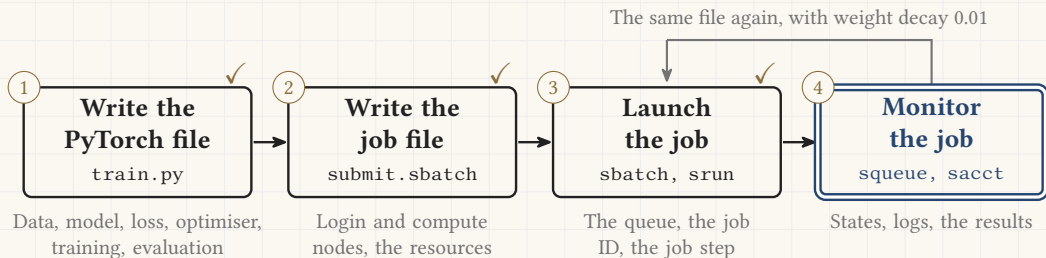
Step 4 of 4

Monitor the job



Step 4 of 4

Monitor the job



squeue shows where the job stands

While the job is in the queue, squeue shows its state; scontrol explains a wait.

```
ooo
```

```
$ squeue -j "$job1" -o '%.18i %.9T %.10M %.6D %R'
```

squeue shows where the job stands

While the job is in the queue, squeue shows its state; scontrol explains a wait.

```
ooo
```

```
$ squeue -j "$job1" -o '%.18i %.9T %.10M %.6D %R'
```

PENDING



Waiting, with a reason such
as Resources or Priority

squeue shows where the job stands

While the job is in the queue, `squeue` shows its state; `scontrol` explains a wait.

```
ooo
```

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$ squeue -j "$job1" -o '%.18i %.9T %.10M %.6D %R'
```

PENDING



Waiting, with a reason such
as Resources or Priority

RUNNING



On a compute
node; the log grows

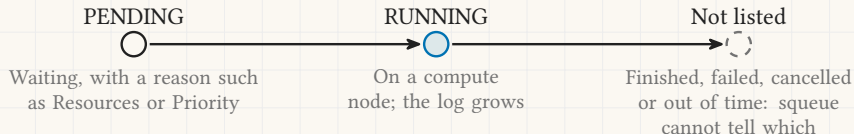


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queue shows where the job stands

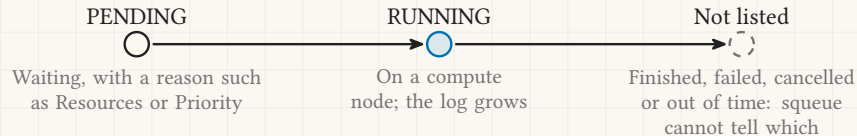
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```
ooo
```

```
$ squeue -j "$job1" -o '%.18i %.9T %.10M %.6D %R'
```

```
$ scontrol show job "$job1"
```

Every detail of one job, the reason for a wait among them. Pending past two minutes: ask, do not resubmit.



Gone from the queue is not success

The logs and sacct say how the job ended.

ooo

```
$ tail -n 20 "logs/cx-pytorch-${job1}.out"
```

What `train.py` printed, epoch by epoch

Gone from the queue is not success

The logs and sacct say how the job ended.

ooo

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$ tail -n 20 "logs/cx-pytorch-${job1}.out"
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What `train.py` printed, epoch by epoch

```
$ tail -n 20 "logs/cx-pytorch-${job1}.err"
```

Errors and warnings

Gone from the queue is not success

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ooo

```
$ tail -n 20 "logs/cx-pytorch-${job1}.out"      What train.py printed, epoch by epoch
$ tail -n 20 "logs/cx-pytorch-${job1}.err"      Errors and warnings
$ sacct -j "$job1" --format=JobID,State,ExitCode,Elapsed,AllocTRES
```

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- COMPLETED, exit code 0:0
And summary.json exists

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FAILED, TIMEOUT or CANCELLED

Read the .err log first

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$ tail -n 20 "logs/cx-pytorch-${job1}.err"      Errors and warnings
$ sacct -j "$job1" --format=JobID,State,ExitCode,Elapsed,AllocTRES
$ scancel <JOB_ID>    Cancels a job by its number, waiting or running
```



COMPLETED, exit code 0:0

And summary.json exists



FAILED, TIMEOUT or CANCELLED

Read the .err log first

What the job leaves behind

Every file carries the job's number, so two runs never mix.

`logs/cx-pytorch-<JOB_ID>.out` The epochs, and the devices of the first step

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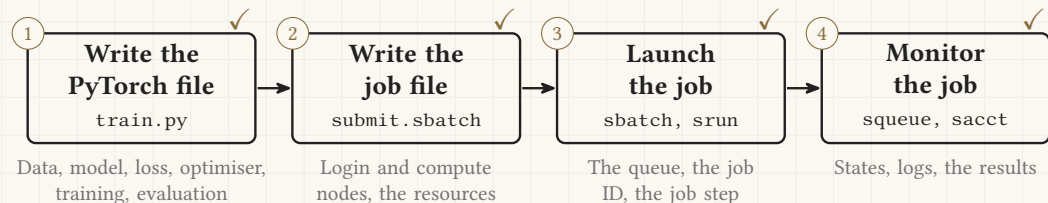
<code>logs/cx-pytorch-<JOB_ID>.out</code>	The epochs, and the devices of the first step
<code>logs/cx-pytorch-<JOB_ID>.err</code>	Errors and warnings
<code>runs/<JOB_ID>/config.json</code>	Settings, seeds, versions, host and GPU
<code>runs/<JOB_ID>/metrics.csv</code>	Train and validation loss and accuracy, every epoch
<code>runs/<JOB_ID>/summary.json</code>	Written only after a successful finish

The `FIRST_STEP` line of the log: where the first step's five tensors were

model cuda:0	features cuda:0	labels cuda:0	logits cuda:0	loss cuda:0
-----------------	--------------------	------------------	------------------	----------------

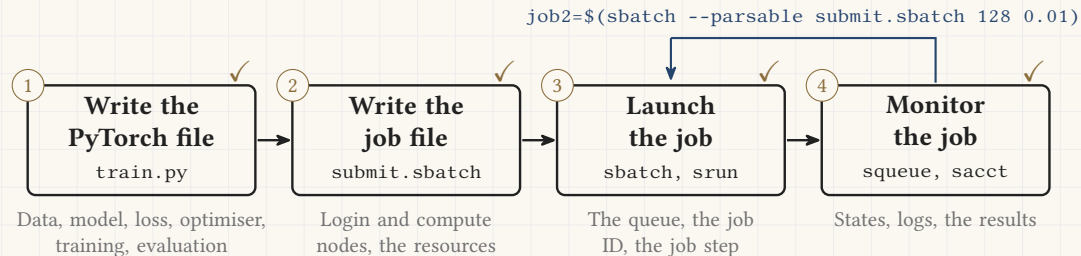
Once more, with weight decay

The same file and the same job file, one number changed. Two folders to compare.



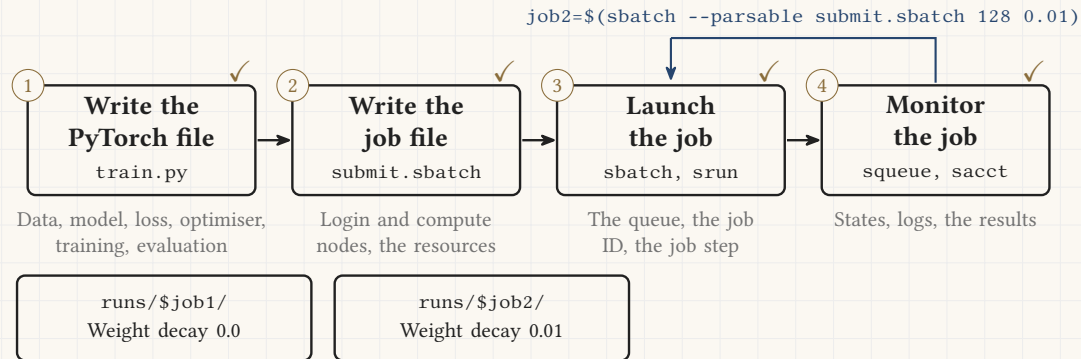
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